

Yes. There is a **good idea inside this proposal**, but I would not implement the five steps exactly as written. As a strict examiner, I would separate them into:

- **strong and defensible:** operating-regime gating, correct RQ3 framing, abstention/no-candidate output;
- **promising but needs technical discipline:** sum/difference residual coordinates;
- **high-risk if presented as originally planned science:** rewriting the FMEA rules after seeing results and using synthetic faults to claim validation.

The biggest technical correction is this:

Sum and difference do not create new information. They re-express the same two residuals in a coordinate system that separates common thermal behaviour from differential thermal behaviour.

That can be very useful, but it is not a magical cure for the ($r \approx 0.95$) problem.

1. My overall verdict

I would score the proposed package like this:

Proposal	Examiner verdict
Reframe RQ3 around plausibility, not confirmed diagnosis	Strong — do it
Gate interpretation to the NBM's valid operating regime	Strong — probably do it, with formal methodological ruling
Transform oil/bearing residuals into common + differential components	Technically promising — worth investigating
Rewrite FMEA rules around those components	Potentially defensible, but currently too post-hoc
Synthetic planted faults	Useful only as rule/implementation verification; not diagnostic validation
Claim “works across 82% of monitoring”	Too strong as written
Treat EVENT-001 abstention as evidence of honest behaviour	Yes, but not proof that the interpretation was correct

My preferred solution is therefore **not to throw this away**, but to modify the plan considerably.

2. The sum/difference idea is technically legitimate — but understand what it actually does

Let your normalised thermal residuals be:

[
 $z_o = \text{oil residual}$
]

[
 $z_b = \text{bearing residual}$
]

Instead of working directly with (z_o) and (z_b) , define:

[
 $C = \frac{z_o + z_b}{\sqrt{2}}$
]

and

[
 $D = \frac{z_b - z_o}{\sqrt{2}}$
]

I would call these:

- (C): **common-mode thermal residual**
- (D): **differential thermal residual**

rather than simply “sum” and “difference”.

What (C) means

If both residuals rise together:

[
 $z_o \uparrow, \quad z_b \uparrow$
]

then:

[
 $C \uparrow$
]

So (C) answers approximately:

“Is the monitored drivetrain thermal system collectively hotter than expected?”

What (D) means

If bearing residual rises more strongly than oil:

$$\left[\begin{array}{l} z_b > z_o \end{array} \right]$$

then:

$$\left[\begin{array}{l} D > 0 \end{array} \right]$$

If oil rises more strongly:

$$\left[\begin{array}{l} z_o > z_b \end{array} \right]$$

then:

$$\left[\begin{array}{l} D < 0 \end{array} \right]$$

So (D) asks:

“Is the bearing unusually hot relative to the oil, or vice versa?”

That is genuinely useful for interpretation.

3. But the transform does NOT discover extra information

This is critical.

The transformation is invertible:

$$\left[\begin{array}{l} z_b = \frac{C+D}{\sqrt{2}} \end{array} \right]$$

$$[z_o = \frac{C-D}{\sqrt{2}}]$$

Therefore:

$$[(z_o, z_b) \leftrightarrow (C, D)]$$

contains **exactly the same information**.

You have not created another sensor.

You have rotated the coordinate system.

That is still valuable because your FMEA reasoning may be much easier in these coordinates.

Instead of asking:

oil high? bearing high? which one was slightly higher?

you can ask directly:

common heating or differential heating?

That is cleaner.

4. Why high correlation can actually make the differential residual interesting

You measured approximately:

$$[r(z_o, z_b) \approx 0.93-0.95]$$

across the partitions.

Suppose both residuals are standardised to approximately unit variance.

Then:

[
Var(C)\approx 1+\rho
]

while:

[
Var(D)\approx 1-\rho
]

If:

[
\rho=0.95
]

then roughly:

[
Var(C)\approx 1.95
]

but:

[
Var(D)\approx 0.05
]

This means normal operation has relatively little **oil-versus-bearing differential variation**.

That can potentially be useful.

A local bearing-specific deviation could appear clearly in (D) even when both raw residuals broadly move together.

For example:

Normal:

[
z_o=1.8,\quad z_b=1.9
]

then:

[
D\approx 0.07
]

But if the bearing develops extra heating:

[
 $z_o=1.8, \text{ } \text{ } z_b=3.4$
]

then:

[
 $D \approx 1.13$
]

Now the bearing-specific departure is obvious.

So the idea is technically sensible.

5. However, there is a mathematical trap

If you standardise the two residuals to equal variance first, then the sum/difference coordinates can be approximately orthogonal **by construction**.

Therefore, if today's finding is simply:

“C and D are uncorrelated in the healthy operating regime,”

that by itself is **not strong evidence of diagnostic discrimination**.

It may partly be algebra.

What you need to demonstrate is something stronger:

Does (D) contain stable, meaningful differential behaviour that corresponds to physically interpretable departures?

Not merely:

[
 $\text{Corr}(C,D) \approx 0.$
]

So before making this central to the thesis, I would want to see:

- variance of (C) and (D);
- distributions by operating regime;
- temporal behaviour;
- sensitivity to normal operating variables;
- stability across turbines;
- whether meaningful differential excursions actually occur.

The current trajectory audit does **not** yet contain the new sum/difference result. It establishes the high original residual correlation and the 82.1%/17.9% operating-regime split, but not that the new transformation “works perfectly”.

So today's result needs a proper recorded artefact before it becomes thesis evidence.

6. The 82% finding is useful — but phrase it correctly

Your audit shows approximately:

[
82.1%
]

of monitoring observations are inside the fitted operating regime, while:

[
17.9%
]

are outside it. The out-of-regime rows have dramatically worse residual behaviour.

Therefore the statement:

“the trick works across 82% of monitoring”

is currently too strong.

What the existing evidence supports is:

“Approximately 82.1% of monitoring observations lie within the operating regime represented by the healthy NBM.”

If the new (C,D) analysis confirms useful behaviour there, you might eventually say:

“The common/differential interpretation was restricted to the 82.1% of monitoring observations lying within the NBM's fitted operating regime.”

But:

[
82.1%\ \text{eligible}
]

does not automatically mean:

[
82.1%\ \text{successfully interpretable}.
]

Keep those quantities separate.

7. “Only let the diagnosis layer speak during normal operation” — I mostly agree

This is one of the best parts of the proposal.

Your audit showed that outside the fitted regime, the NBM is producing large systematic residual errors:

[
\text{out-of-regime bearing bias}\approx-11.75^\circ\text{C}
]

and the exceedance rate jumps dramatically.

A residual is only diagnostically meaningful if:

[
 $r=y-\hat{y}$
]

represents:

deviation from a trustworthy healthy prediction.

If $(\hat{y})$ itself is invalid because the model is extrapolating, then a large (r) cannot safely be interpreted as degradation.

Therefore I agree with the principle:

[
\boxed{
\text{No physical interpretation outside model support}
}
]

This is good engineering.

But don't call the 17.9% “meaningless data”

The observations themselves are not meaningless.

The **NBM residual interpretation is unreliable there.**

That is the correct language.

8. Should this gate apply to detection or only diagnosis?

This is important because you do not want to accidentally rewrite RQ2.

I recommend the safer formulation:

RQ2

Keep the original results and the negative finding.

Do **not** use the new transformation to rescue RQ2.

RQ3 interpretation layer

Add an eligibility condition:

```
[  
\text{Interpretation eligible}  
\iff  
\text{observation lies within NBM operating support}.  
]
```

Then outside support:

Interpretation withheld — outside NBM operating domain.

That is much safer than assigning a physical candidate mechanism to an extrapolation artefact.

It is also consistent with the audit's recommendation that operating-state gating requires an explicit methodological ruling.

9. The RQ3 reframing is strong

This part of the proposal is basically correct.

Your audit itself phrases RQ3 around whether FMEA-informed interpretation produces **plausible diagnostic output**, not whether it recovers maintenance-confirmed truth.

You should distinguish:

Question A

Can the framework assign physically plausible candidate interpretations?

versus

Question B

Is the assigned mechanism actually the true fault?

These are not equivalent.

Your data cannot answer B.

But a carefully constructed system can still provide evidence about A.

The key outputs can be:

[
{\text{candidate mechanism(s)}}
]

or

[
{\text{no supported candidate}}.
]

That is a legitimate explainability architecture.

10. EVENT-001: “I don't know” is useful, but don't call it proven correct

This proposal slightly overstates the case here.

EVENT-001 is a lubrication/filter restriction whose direct published indicators are pressure/filter information rather than the two thermal residuals.

So having the thermal interpretation layer decline to assign one of the positive thermal mechanisms is **reasonable behaviour**.

You can say:

“The interpretation layer did not force a positive thermal mechanism onto an event whose documented primary indicators lie outside the modelled signal set.”

That is good.

But I would **not** say:

“The system correctly said ‘I don't know’.”

Why?

Because “correctly” still requires independent truth about what the appropriate output should have been.

Safer:

“The system abstained from unsupported positive attribution.”

That is demonstrably true.

And it is a strength.

11. The biggest issue: rewriting the rules now

This is where I become cautious.

The proposed new rules are approximately:

common heat → cooling/lubrication/load
bearing differential → bearing fault
oil differential → oil-side fault

Conceptually, excellent.

As final FMEA rules, **too crude**.

For example:

[
D>0
]

does not automatically mean:

bearing fault.

It could also arise from:

- bearing-channel sensor bias;
- different thermal lag;
- local ambient effects;
- model error specific to the bearing target;
- operating transitions;
- other local thermal sources.

Similarly:

[
C>0
]

does not uniquely distinguish:

- gear loss;
- cooling impairment;
- lubrication issues;
- general load-related thermal mismatch.

Therefore don't throw away the existing physics-informed conditions.

12. My preferred solution: use C and D to IMPLEMENT the existing rule logic, not replace it

This is the strongest version of your idea.

Keep the FMEA architecture.

But express its signatures more cleanly in common/differential coordinates.

For example, conceptually:

R1 — oil-led mechanism

Early:

[
D<0
]

because oil is elevated relative to bearing.

Then later:

```
[  
C>0  
]
```

as both streams become elevated.

Plus the existing load-dependence requirement.

R2 — bearing-led mechanism

Early:

```
[  
D>0  
]
```

because bearing is elevated relative to oil.

Later:

```
[  
C>0  
]
```

as oil follows.

R4 — circuit-wide/common thermal elevation

```
[  
C\gg0  
]
```

while:

```
[  
D\approx0  
]
```

because both move together.

That is a very elegant way of expressing what your original rules were trying to encode.

The key difference is:

You are not inventing new mechanisms from C and D. You are using C and D as derived coordinates to operationalise the existing FMEA signatures.

That is much easier to defend.

13. Is that still a methodological change?

Yes.

Do not pretend otherwise.

Even though:

$$\begin{bmatrix} (z_o, z_b) \end{bmatrix} \rightarrow (C, D)$$

is information-preserving, introducing (C,D) into the formal interpretation engine after inspecting the results is a **methodological amendment**.

The fact that you “add the new rules alongside the old rules” does **not** remove the post-hoc problem.

Timing determines post-hoc status, not whether the original code remains available.

An examiner can still ask:

“You saw the original rule system fail, then created a new representation that produced more favourable discrimination. Why isn't this result-driven method development?”

You need a clean answer.

14. How to make the C/D amendment defensible

I would require four conditions.

1. Record the discovery transparently

Create an ADR stating:

- original residual correlation problem;
- why the original state representation was inadequate;

- mathematical definition of (C,D);
- that the transformation was introduced **after the initial RQ3 implementation exposed a discrimination limitation.**

Do not hide chronology.

2. Freeze the transformation before evaluating rule outcomes

Specify exactly:

$$\begin{bmatrix} C=(z_o+z_b)/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} D=(z_b-z_o)/\sqrt{2}. \end{bmatrix}$$

Specify thresholds, persistence, timing and operating eligibility before running the new rule evaluation.

3. Do not delete the original result

Chapter 4 should show:

Original interpretation architecture → discrimination limitation.

Then:

Remedial/common-differential analysis → exploratory or amended interpretation analysis.

That transparency substantially reduces the “moving goalposts” criticism.

4. Downgrade the inferential status

Unless you have untouched data remaining, I would label the C/D result:

post-hoc methodological refinement / exploratory robustness analysis

rather than pretending it was the original confirmatory RQ3 test.

That is still perfectly worthwhile in an MSc.

15. The synthetic-fault proposal is the most dangerous part

I would **not use synthetic planted faults to claim diagnostic validation.**

Suppose you construct a synthetic bearing event by doing:

$$\left[\begin{array}{l} z_b \leftarrow z_b + \delta(t) \end{array} \right]$$

while keeping oil approximately unchanged.

Then your differential becomes:

$$\left[\begin{array}{l} D = \frac{z_b - z_o}{\sqrt{2}} > 0 \end{array} \right]$$

You have designed the injected anomaly specifically to satisfy the bearing rule.

If the rule engine then says:

bearing-type candidate,

that proves primarily that the code implements the rule you wrote.

It does **not** prove that real bearing faults create that pattern.

It can become almost tautological.

16. What synthetic injection CAN legitimately prove

It can test:

Software correctness

Does the intended rule fire when its defining conditions exist?

Structural identifiability

Can the rule system distinguish idealised common-mode from differential-mode patterns?

Sensitivity

At what perturbation magnitude/persistence does each rule become detectable?

Cross-talk

Does a simulated bearing-differential signal incorrectly activate the common-mode rule?

Those are valuable tests.

But name them appropriately:

synthetic rule-verification experiments

or

controlled signature-injection tests

not:

“diagnostic validation.”

17. I would not write “accepted standard method” without evidence

The proposal says:

“This is an accepted, standard method in the field.”

Do not put that in the dissertation unless one of your literature sources explicitly supports synthetic fault/signature injection as a validation approach for this class of SCADA diagnosis.

The current trajectory audit provides **no evidence for that statement**.

So either:

- find a source already in your literature base supporting it; or
- describe it neutrally as a controlled verification experiment.

Do not assert methodological consensus without evidence.

18. Synthetic tests also do not solve your real RQ3 limitation

Even perfect synthetic results:

[
100%\ \text{correct rule recovery}
]

would not establish:

[
\text{diagnostic accuracy on real faults}=100%.
]

Your real limitation remains:

[
n_{\text{maintenance-confirmed faults}}=0.
]

Synthetic data cannot manufacture real-world ground truth.

So RQ3 still cannot become:

“validated diagnosis.”

It can become something more defensible:

“The interpretation architecture was structurally verified under controlled signature injections and descriptively evaluated on the available real event; real-world diagnostic correctness remains unvalidated.”

That would be a respectable outcome.

19. What I think the strongest revised RQ3 package looks like

I would use this hierarchy:

Primary evidence

Real data, in operating support

Analyse the common and differential residual coordinates during eligible monitoring operation.

Structural FMEA analysis

Map existing candidate mechanisms into:

[
(C,D,\text{timing},\text{persistence},\text{load dependence})
]

rather than replacing the physics with only (C,D).

Abstention

Allow:

[
R5=\text{no gearbox-consistent thermal candidate}
]

and:

[
R_{\text{OOD}}=\text{interpretation withheld outside model support}
]

Conceptually those are different:

- R5: valid model domain, but observed pattern fits no gearbox rule;
- OOD/withheld: residual itself is not trustworthy enough to interpret.

That distinction would be technically excellent.

Optional synthetic verification

Use idealised injections only to verify that the implementation can distinguish predefined signature classes.

Do not use it as empirical diagnostic validation.

20. This gives you a much better output vocabulary

Instead of forcing:

fault = R1/R2/R3/R4

your system can produce four different kinds of answer.

Type A — positive candidate

“Bearing-led thermal pattern; R2-consistent.”

Type B — ambiguous candidates

“Oil-led pattern consistent with R1/R3; available thermal evidence cannot discriminate further.”

Type C — valid-domain abstention

“No gearbox-consistent thermal candidate.”

Type D — model-domain abstention

“Interpretation withheld: operating condition outside the NBM's fitted support.”

That is a sophisticated interpretation framework.

It knows:

- when it has evidence;
- when evidence is ambiguous;
- when nothing fits;
- when it should not interpret the residual at all.

That is substantially more defensible than trying to force one mechanism per alert.

21. Does this rescue RQ3?

Potentially yes, as an **architecture/capability question**.

Not as validated fault diagnosis.

If your official RQ3 is essentially:

Can coordinated residual behaviour support physically plausible FMEA-informed interpretation?

then a defensible positive result could be:

Yes, within the NBM's supported operating regime, the two residual streams can be decomposed into common and differential thermal components that make distinct classes of coordinated behaviour representable within the FMEA interpretation layer. The framework can return candidate mechanisms, retain ambiguity where signatures overlap, and abstain where the available thermal evidence is insufficient. These outputs are plausibility-based and are not validated against maintenance-confirmed fault mechanisms.

That is substantially stronger than:

“RQ3 failed.”

But only if the common/differential analysis genuinely supports the claim after the rules are frozen.

22. Does this rescue RQ2?

No, and don't try.

RQ2 should remain your negative finding.

The original RQ2 deals with whether coordination improves **detection** compared with single-signal monitoring.

The common/differential decomposition is primarily an **interpretation representation**.

Keep those separate.

Then your thesis becomes intellectually interesting:

[
\boxed{\text{RQ1: positive}}
]

NBM provides better healthy modelling.

[
\boxed{\text{RQ2: negative}}
]

Two highly correlated thermal residuals do not improve detection simply through coincidence-based coordination.

[
\boxed{\text{RQ3: bounded positive/partial}}
]

The same two signals may still carry interpretable **relative structure** when expressed as common versus differential thermal behaviour, even though they did not improve detection.

That is actually a coherent research story.

23. This distinction is scientifically important

High correlation does not imply:

“the second signal contains absolutely nothing useful.”

It means much of the variation is shared.

You can conceptually decompose:

$$z_o = \text{common behaviour} + \text{oil-specific part}$$
$$z_b = \text{common behaviour} + \text{bearing-specific part}.$$

RQ2's simple 1-of-2 versus 2-of-2 logic mostly sees the common behaviour.

The differential:

$$D = z_b - z_o$$

is designed specifically to cancel the common component and expose the relative component.

So it is **quite possible** for:

coordination to add little to binary detection,

while:

relative residual structure still adds interpretive information.

That is not contradictory.

It could become one of the strongest intellectual points in the dissertation.

24. My assessment of your five proposed steps

Step	Verdict	Recommended treatment
1. Reframe claim around plausibility	9/10	Do it. Do not claim diagnostic correctness.
2. Formalise C/D in operating regime	8/10	Promising. Record formally, verify across turbines/regimes, don't call it "perfect".
3. Rewrite rules around C/D	5/10 as written; 8/10 if modified	Don't replace rules. Use C/D to operationalise existing FMEA signatures.
4. Synthetic planted faults	4/10 if called validation; 7/10 as supplementary rule verification	Keep clearly synthetic and secondary.
5. Final positive claim	6/10 as written; 9/10 after narrowing	Replace "works across 82%" and "names the right cause" with precise capability claims.

25. My recommended solution

I would **not** execute the proposed plan exactly.

I would use this version:

1. **Keep RQ1 and RQ2 unchanged.** RQ2 remains a bounded negative result.
 2. **Formally introduce an RQ3-only operating-eligibility condition** based on the existing 50 kW model-support boundary. Do not interpret out-of-support residuals.
 3. **Introduce common-mode and differential residual coordinates as an algebraic transformation of the same two normalised residual streams**, not as new sensors or a new NBM.
 4. **Do not invent a completely new FMEA taxonomy.** Rewrite the implementation of the existing rules using (C), (D), temporal order, persistence and load dependence.
 5. **Retain the old RQ3 implementation and its adverse result in the audit trail.** Explicitly label the C/D version as a methodological refinement prompted by the measured collinearity problem.
 6. **If you run synthetic injections, treat them as supplementary structural verification only.** Never use them to claim real diagnostic accuracy.
 7. **Use EVENT-001 as a real-data abstention case**, not as proof that a mechanism was correctly diagnosed.
 8. **Keep the final limitation:** maintenance-confirmed diagnostic correctness remains unavailable.
-

The final claim I would be comfortable examining

Something close to:

The two thermal residuals did not provide a consistent advantage for coordinated anomaly detection because their dominant variation was highly correlated. However, within the NBM's supported operating regime, their common and differential components provided a more informative representation for FMEA-based interpretation: common-mode behaviour represented shared thermal deviation, while differential behaviour represented relative oil–bearing deviation. The interpretation layer could therefore express candidate mechanism classes, retain ambiguity where the thermal signatures were not discriminable, and abstain where no supported thermal interpretation was available. Controlled synthetic signature tests, if included, verify the rule implementation rather than real-world diagnostic correctness; maintenance-confirmed diagnostic validation remains outside the available ground truth.

That is a thesis I would take seriously.

The **sum/difference idea is worth pursuing**, but the safe way to use it is as an **interpretable reparameterisation of your existing two residuals**, not as a late invention that magically “proves” RQ3.